# MAS508-G Control Theory
# Lecture 3: State Space Design

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# L3. State Space Design

a) Time Response
b) Controllability
c) Observability
d) Pole Placement Control
e) Observer Design
f) Output Feedback Control
g) Optimal Control - Linear Quadratic Control (LQR)
h) Optimal Control – LQG

Textbook, “Control Systems Engineering”, Chapters 4 &12

# L3e. Observer Design

Book “Control Systems Engineering”, Chapter 12.5, 12.7

# Observer

- State feedback assume that *all the state variables are available* for feedback control.
- In practice, this assumption is *not* always met either because not all the state variables can be *directly measured* or *the number of measurement devices is limited* due to cost consideration.
- We will discuss the problem of reconstructing the state vector by using *available inputs and outputs* of the plant (system to be controlled) to drive a device called a *state observer*.
- System is observable.

# Open-Loop Observer

- Original System

$$\dot{\mathrm{x}} = \mathrm{A}\mathrm{x} + B u$$
$$y = C\mathrm{x}$$

- Open-loop Observer

If $\{\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}\}$ are known, one can simulate a model with an accessible state vector for estimating $\boldsymbol{x}$.

$$\dot{\hat{\mathrm{x}}} = \mathrm{A}\hat{\mathrm{x}} + B\, u$$
$$\hat{y} = C\hat{\mathrm{x}}$$

where $\hat{\boldsymbol{x}}$ denotes an estimate of $\boldsymbol{x}$.

# Open-Loop Observer

Let $\boldsymbol{e_x}$ be the error in our estimate so that

$$e_x = \mathrm{x} - \hat{\mathrm{x}}$$

The derivative of estimation error

$$\begin{aligned}\dot{e}_x &= \dot{\mathrm{x}} - \dot{\hat{\mathrm{x}}} = \mathrm{A}\mathrm{x} + B\mathrm{u} - (\mathrm{A}\hat{\mathrm{x}} + B\mathrm{u}) \\ &= \mathrm{A}e_x\end{aligned}$$

The solution of above is

$$e_x(t) = e^{\mathrm{A}t}e_x(0)$$

- $\boldsymbol{If\ e_x(0) = 0}$, the observer system will deliver the estimate of $\boldsymbol{x}$ exactly;

# Open-Loop Observer

$$e_x(t) = e^{At} e_x(0)$$

- *If* $\boldsymbol{e_x(0) \neq 0}$ and *all the eigenvalues* of $\boldsymbol{A}$ have *negative real parts*, $\boldsymbol{e_x(t)}$ decay to zero exponentially. The estimate becomes better as time progresses;

- *If* $\boldsymbol{e_x(0) \neq 0}$ and *some the eigenvalues* of $\boldsymbol{A}$ have *positive real parts*, any small initial error $\boldsymbol{e_x(0)}$ may grow rapidly → *unacceptable*.

# Open-Loop Observer

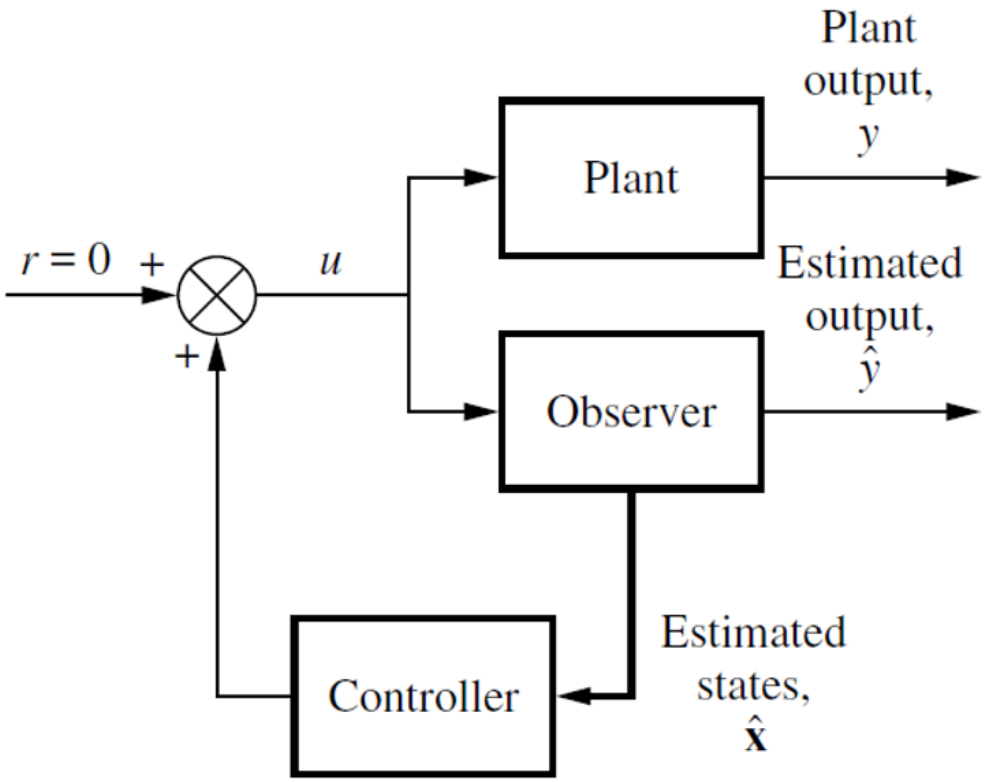
$$\dot{\mathrm{x}} = \mathrm{A}\mathrm{x} + Bu$$
$$y = C\mathrm{x}$$

$$\dot{\hat{\mathrm{x}}} = \mathrm{A}\hat{\mathrm{x}} + B\,u$$
$$\hat{y} = C\hat{\mathrm{x}}$$

$$\dot{e}_x = \dot{\mathrm{x}} - \dot{\hat{\mathrm{x}}} = \mathrm{A}\mathrm{x} + B\mathrm{u} - (\mathrm{A}\hat{\mathrm{x}} + B\mathrm{u}) = \mathrm{A}e_x$$

- The dynamics of the error is unforced and depends on the system matrix A.
- The speed of the convergence between the actual state and estimated state is the same as the transient response of the system itself. (Slow)
- How to increase the speed of convergence? – Closed-loop Observer

# Open-Loop Observer




$$\dot{\hat{\mathrm{x}}} = \mathrm{A}\hat{\mathrm{x}} + B\,u$$
$$\hat{y} = C\hat{\mathrm{x}}$$

# Closed-Loop Observer

- Since both $\boldsymbol{y}$ and $\hat{\boldsymbol{y}}$ are available in the *open-loop* observer, we may compare them and use the *difference* $(\boldsymbol{y} - \hat{\boldsymbol{y}})$ as a *corrective term* to construct an improved state observer.

$$\dot{\hat{\mathrm{x}}} = \mathrm{A}\hat{\mathrm{x}} + Bu + L\,(\mathrm{y} - \hat{\mathrm{y}})$$
$$\hat{y} = C\hat{\mathrm{x}}$$

where $L$ is the constant vector. The original system

$$\dot{\mathrm{x}} = \mathrm{A}\mathrm{x} + Bu$$
$$y = C\mathrm{x}$$

# Closed-Loop Observer

Let $e_x = \mathrm{x} - \hat{\mathrm{x}}$

There is

$$y - \hat{y} = C(x - \hat{x}) = C\, e_x$$
$$\dot{\mathrm{x}} - \dot{\hat{\mathrm{x}}} = A(x - \hat{x}) - L(y - \hat{y}) = (A - LC)\, e_x$$

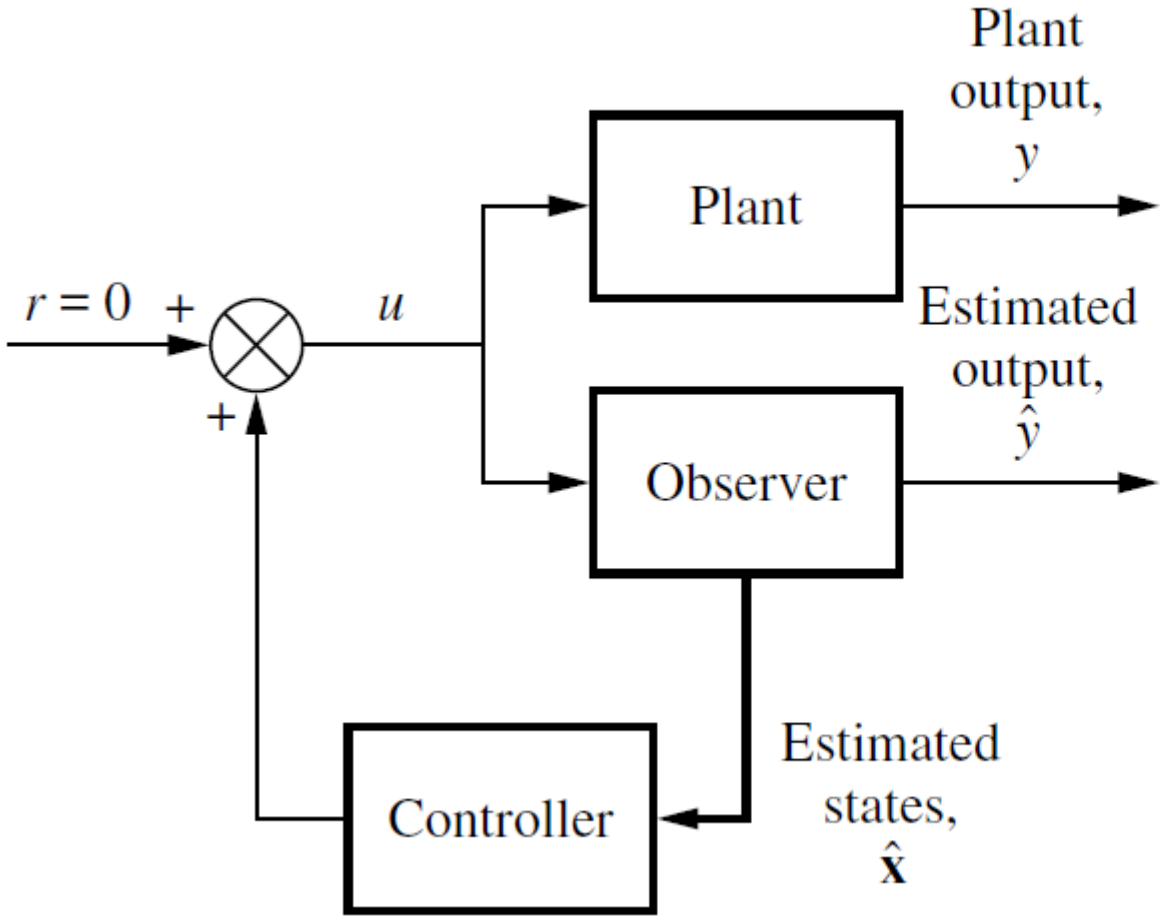
Thus

$$\dot{e}_x = (\mathrm{A} - LC)e_x$$
$$y - \hat{y} = Ce_x$$

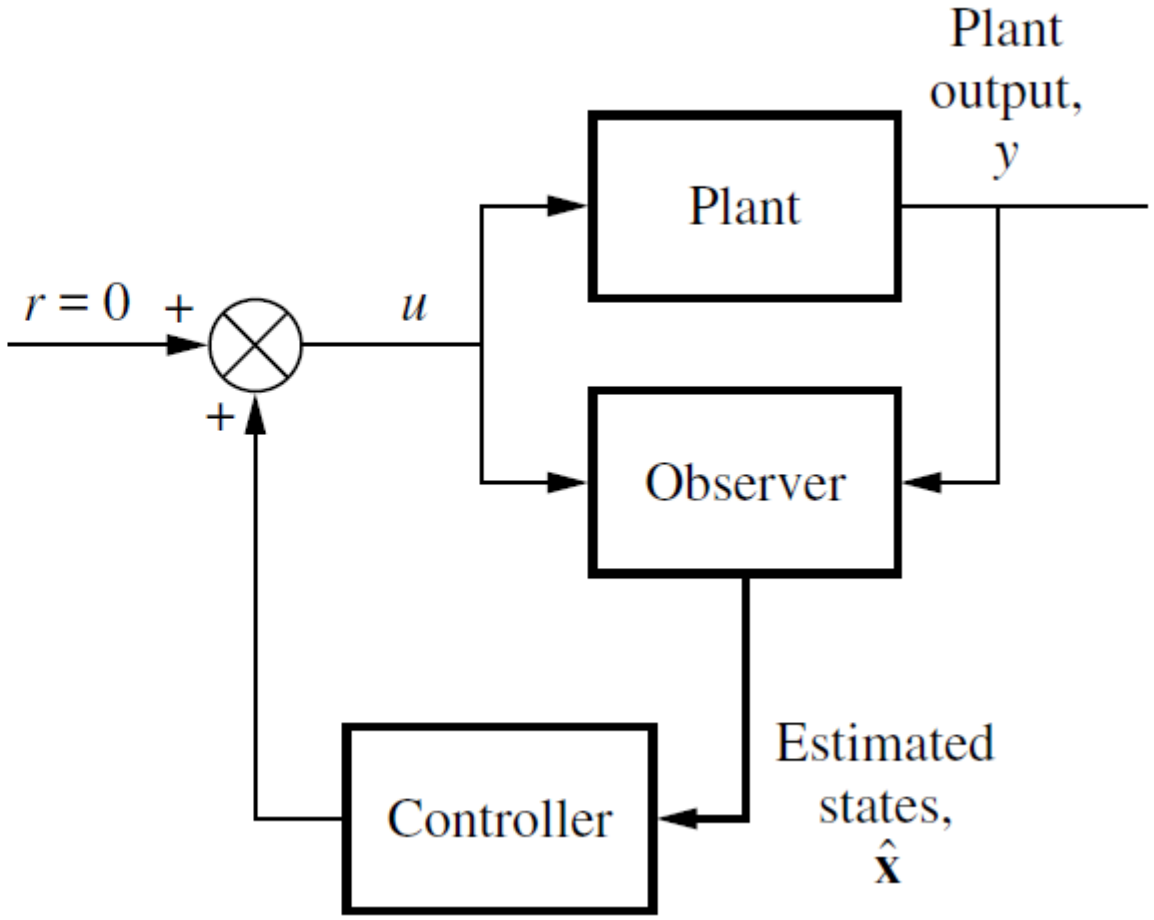
- If *the eigenvalues* of $(\boldsymbol{A - LC})$ can be chosen arbitrarily, by choosing an appropriate $\boldsymbol{L}$, the convergence of $\boldsymbol{e_x}$ and the *convergence speed* can be guaranteed.

# Observer

- ***Open-loop Observer***

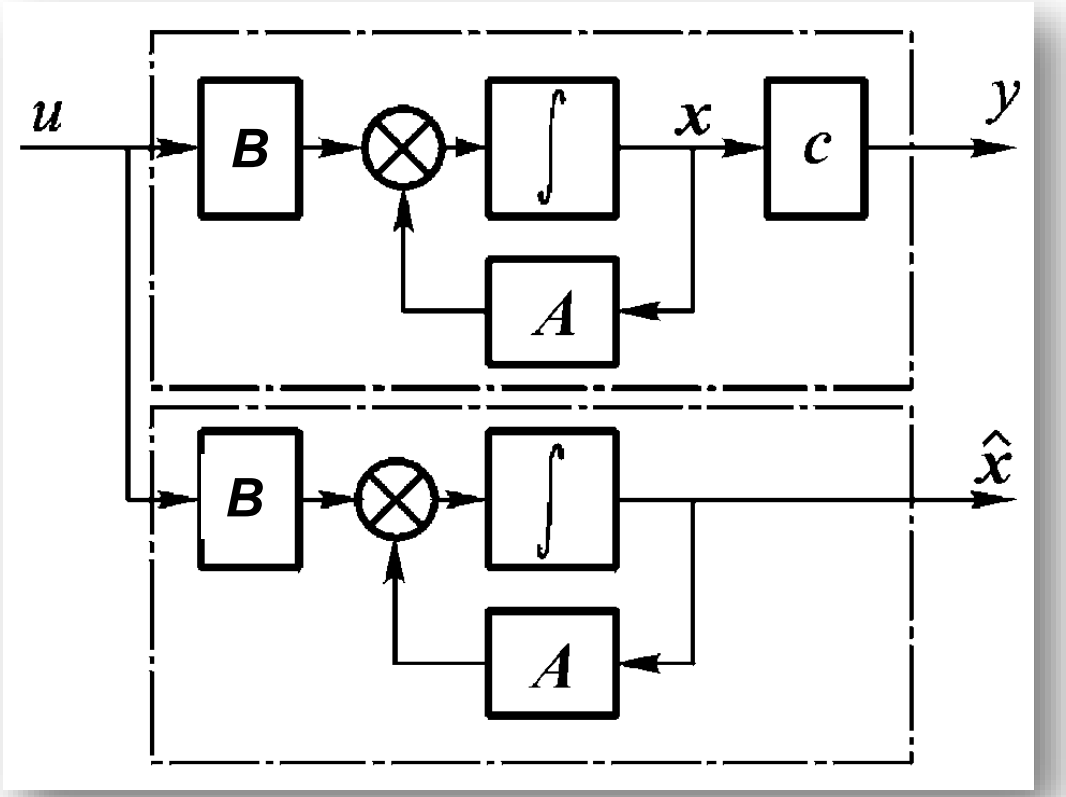



- ***Closed-loop Observer***

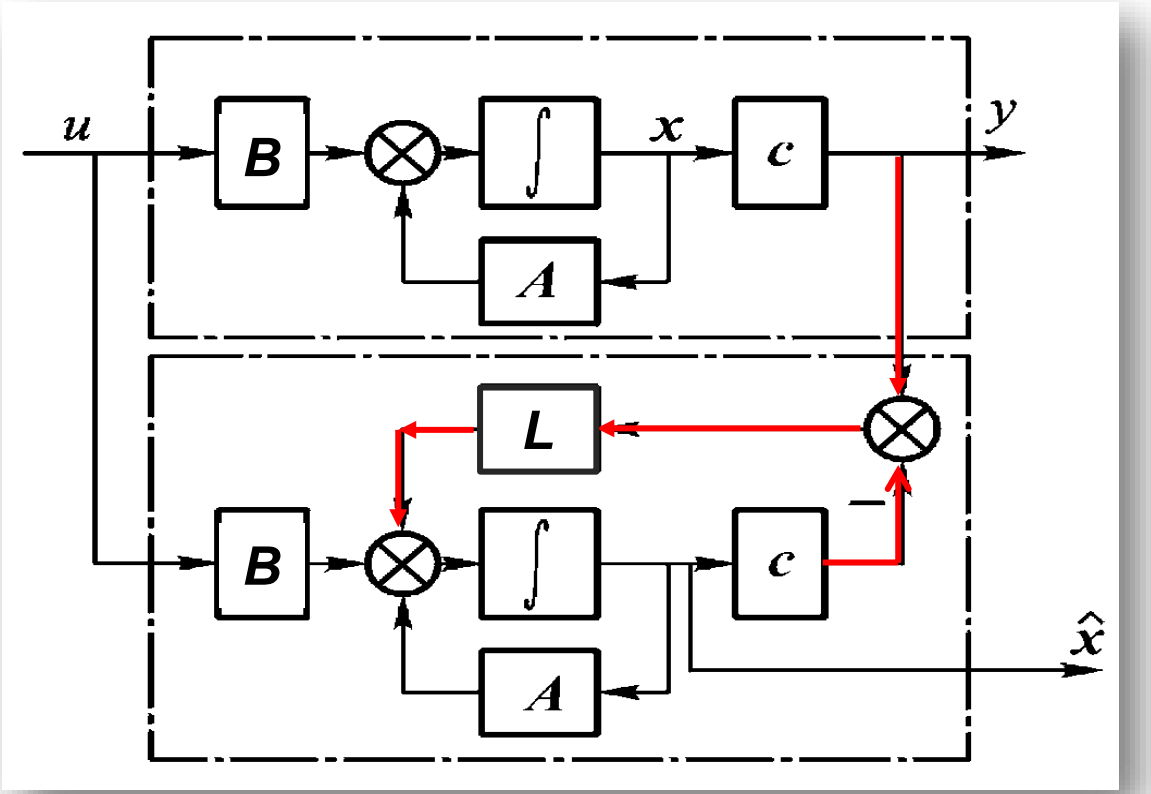


# Observer

- *Open-loop Observer*




- *Closed-loop Observer*



# Closed-loop Observer

$$\dot{\mathrm{x}} = \mathrm{A}\mathrm{x} + Bu$$
$$y = C\mathrm{x}$$

- Theorem: A state observer is described by

$$\dot{\hat{\boldsymbol{x}}} = (\boldsymbol{A} - \boldsymbol{LC})\hat{\boldsymbol{x}} + \boldsymbol{Bu} + \boldsymbol{Ly}$$

can be constructed with *arbitrary eigenvalues* if and only if the system $(\boldsymbol{A},\ \boldsymbol{C})$ is *observable*.

- Method to find $\boldsymbol{L}$ using pole placement: to match the coefficients of $\mathbf{det}\big(\boldsymbol{sI} - (\boldsymbol{A} - \boldsymbol{LC})\big)$ with the coefficients of the desired characteristic polynomial.

# Example. Observer design

- Consider the following system

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$
$$y = \begin{bmatrix} 2 & 0 \end{bmatrix} x$$

Design a state observer, where the eigenvalues of $(\boldsymbol{A} - \boldsymbol{LC})$ are required to be $\lambda_1 = \lambda_2 = -10$.

# Example. Observer design

- Solution. The system is observable, $\text{rank}\begin{bmatrix} C \\ CA \end{bmatrix} = 2$.

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}, C = [2 \quad 0], \text{L} = \begin{bmatrix} l_1 \\ l_2 \end{bmatrix}$$

- Let $L = [l_1, l_2]^T$. The characteristic polynomial of $(\boldsymbol{A - LC})$ is

$$\det[s\text{I} - (\text{A} - LC)] = \det\begin{bmatrix} s + 2l_1 & -1 \\ 2 + 2l_2 & s + 3 \end{bmatrix} = s^2 + (2l_1 + 3)s + 6l_1 + 2l_2 + 2$$

- The required characteristic polynomial of $(\boldsymbol{A - LC})$ is

$$(s+10)^2 = s^2 + 20s + 100$$

$$(2l_1 + 3) = 20$$
$$6l_1 + 2l_2 + 2 = 100$$

- Compute $l_1, l_2$ $\quad l_1 = 8.5, l_2 = 23.5$

# Example. Observer design

- The state observer is designed as

$$\begin{aligned}\dot{\hat{\mathrm{x}}} &= (\mathrm{A} - LC)\hat{\mathrm{x}} + Bu + L\mathrm{y} \\ &= \left(\begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} - \begin{bmatrix} 8.5 \\ 23.5 \end{bmatrix} [2 \quad 0]\right)\hat{\mathrm{x}} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u + \begin{bmatrix} 8.5 \\ 23.5 \end{bmatrix} \mathrm{y} \\ &= \begin{bmatrix} -17 & 1 \\ -49 & -3 \end{bmatrix}\hat{\mathrm{x}} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u + \begin{bmatrix} 8.5 \\ 23.5 \end{bmatrix} \mathrm{y}\end{aligned}$$

# Example 12.5 - Nise book

PROBLEM: Design an observer for the plant. The designed observer poles are $\lambda_{1,2} = A = -10 \pm \mathrm{j}20,\ \lambda_3 = -100.$

$$G(s) = \frac{(s+4)}{(s+1)(s+2)(s+5)} = \frac{s+4}{s^3+8s^2+17s+10}$$

Solution: Represent in state-space form

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u = \begin{bmatrix} -8 & 1 & 0 \\ -17 & 0 & 1 \\ -10 & 0 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \\ 4 \end{bmatrix} u$$

$$y = \mathbf{C}\mathbf{x} = [1 \quad 0 \quad 0]\mathbf{x}$$

# Example 12.5 - Nise book

Solution: a) Open-loop observer

$$\dot{\hat{\mathbf{x}}} = \mathbf{A}\hat{\mathbf{x}} + \mathbf{B}u = \begin{bmatrix} -8 & 1 & 0 \\ -17 & 0 & 1 \\ -10 & 0 & 0 \end{bmatrix} \hat{\mathbf{x}} + \begin{bmatrix} 0 \\ 1 \\ 4 \end{bmatrix} u$$

$$\hat{y} = \mathbf{C}\hat{\mathbf{x}} = [1 \quad 0 \quad 0]\hat{\mathbf{x}}$$

# Example 12.5 - Nise book

Solution: a) Closed-loop observer

$$(\mathbf{A}-\mathbf{LC}) = \begin{bmatrix} -(8+l_1) & 1 & 0 \\ -(17+l_2) & 0 & 1 \\ -(10+l_3) & 0 & 0 \end{bmatrix}$$

The characteristic polynomial $s^3 + (8+l_1)s^2 + (17+l_2)s + (10+l_3)$

The desired characteristic polynomial: $(s+100)(s^2+20s+500) = s^3 + 120s^2 + 2500s + 50{,}000$

$$l_1 = 112,\ l_2 = 2483, \text{ and } l_3 = 49{,}990.$$

# Example 12.5 - Nise book

Solution: b) Closed-loop observer

$$\dot{\hat{x}} = (A - LC)\hat{x} + Bu + Ly$$

$$A = \begin{bmatrix} -8 & 1 & 0 \\ -17 & 0 & 1 \\ -10 & 0 & 0 \end{bmatrix}$$

$$B = [0 \quad 1 \quad 4]^T$$

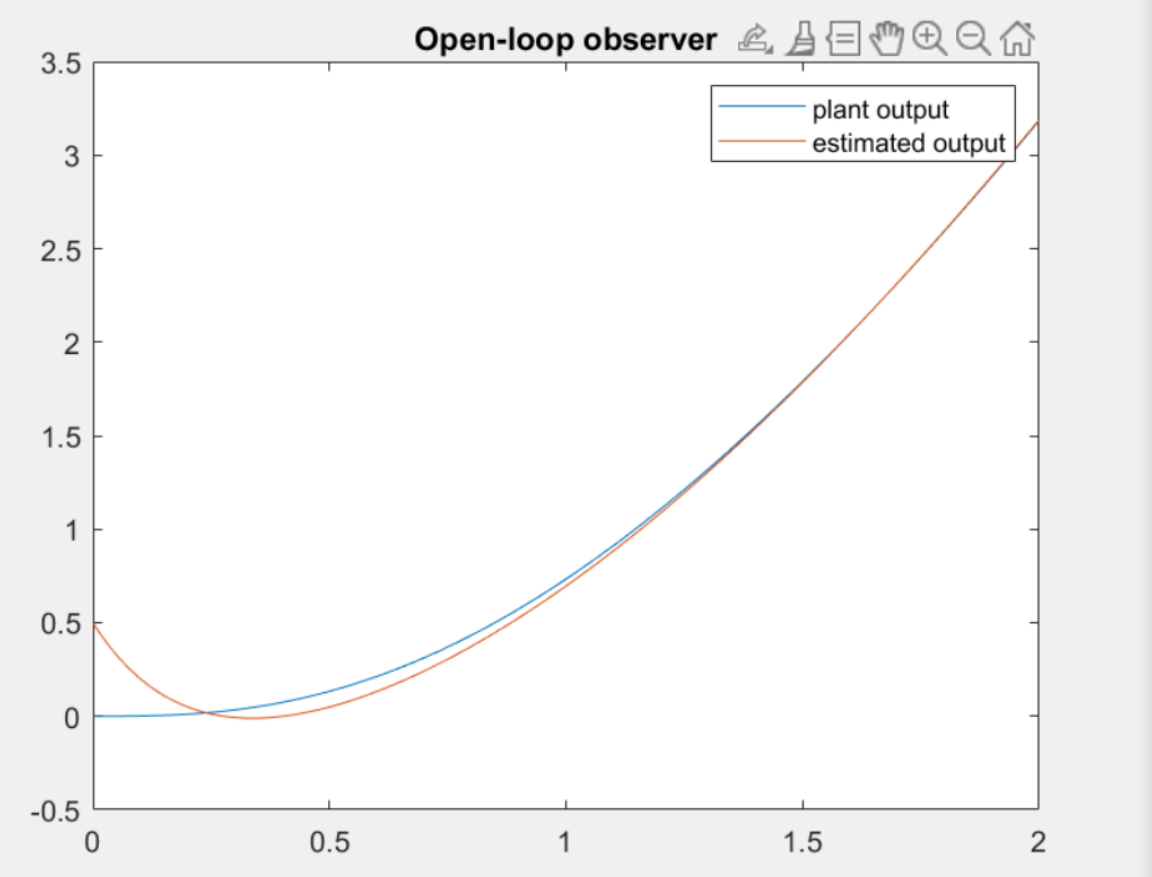
$$C = [1 \quad 0 \quad 0]$$

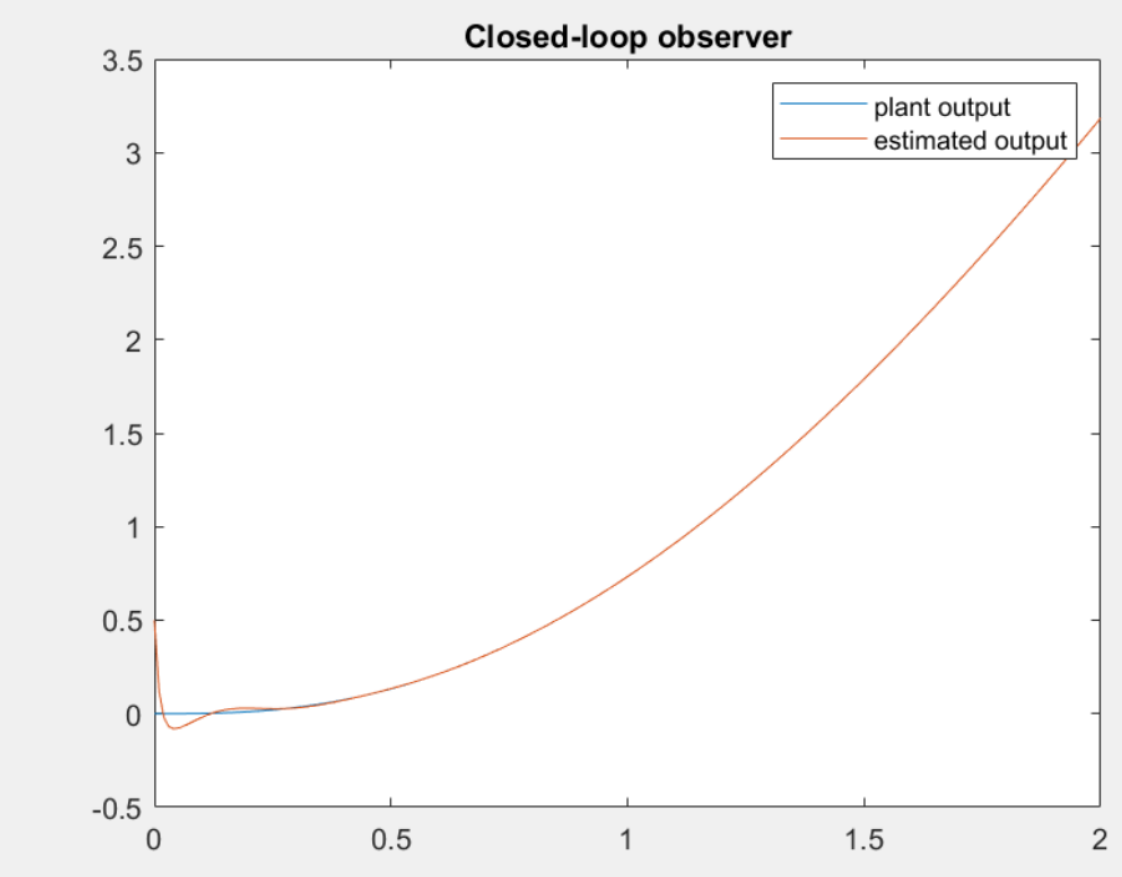
$$L = [112 \quad 2483 \quad 49.990]^T$$

Matlab command: L=place(A',C', P)'

# Example 12.5 - Nise book

Reference $u(t) = 100t$, initial $\mathrm{x}(0) = [0\ 0\ 0]$; initial estimate $\hat{\mathrm{x}}(0) = [0.5; 0; 0]$




Example 12.5 Nise book comparison of Open-loop and Closed-loop observers.

Matlab: L3e_exNise_observer.m
% L=place(A',C', P)';

# Closed-Loop Observer

Plant

$$\dot{\mathrm{x}} = \mathrm{A}\mathrm{x} + Bu$$
$$\mathrm{y} = C\mathrm{x}$$

Closed-loop observer:

$$\dot{\hat{\mathrm{x}}} = (\mathrm{A} - LC)\hat{\mathrm{x}} + Bu + \mathrm{L}\mathrm{y}$$
$$\dot{\hat{\mathrm{x}}} = (A - LC)\,\hat{\mathrm{x}} + Bu + LC\mathrm{x}$$
$$\hat{\mathrm{y}} = C\hat{\mathrm{x}}$$

New system with observer:

$$\begin{bmatrix} \dot{\mathrm{x}} \\ \dot{\hat{\mathrm{x}}} \end{bmatrix} = \begin{bmatrix} A & 0 \\ LC & A - LC \end{bmatrix} \begin{bmatrix} \mathrm{x} \\ \hat{\mathrm{x}} \end{bmatrix} + \begin{bmatrix} B \\ B \end{bmatrix} u$$
$$\begin{bmatrix} \mathrm{y} \\ \hat{\mathrm{y}} \end{bmatrix} = \begin{bmatrix} C & 0 \\ 0 & C \end{bmatrix} \begin{bmatrix} \mathrm{x} \\ \hat{\mathrm{x}} \end{bmatrix}$$

# Takeaways

## Benefits

- If it's observable, then you can design an observer to estimate the states.
- Can use the estimated states in the state feedback control.

## Drawbacks

- Need system dynamic model of system